Structural uniformity of neocortex, revisited

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Contributed by Charles F. Stevens, December 12, 2012 (sent for review June 16, 2012)

The number of neurons under a square millimeter of cortical surface has been reported to be the same across five cortical areas and five species [Rockel et al. (1980) *Brain* 103(2):221–244] despite differences in cortical thickness between the areas. Although the accuracy of this result has been the subject of sharp debate since its publication approximately 30 y ago, the experiments of Rockel et al. have never been directly replicated with modern stereological methods. We have replicated these experiments and confirm the accuracy of the original report. In addition, we have observed that the number of glial cells under a square millimeter of cortical surface depends on cortical thickness, but not on cortical area or species.

allometry | brain size | cell number | evolution | microscopy

A third of a century ago, Rockel et al. (1980), in a paper titled "The basic uniformity in structure of the neocortex" (1), reported a striking observation: the number of neurons underneath a square millimeter of neocortical surface is the same across cortical areas and species, with one exception—primates have approximately twice as many neurons under a square millimeter of their primary visual cortex (area 17) surface compared with other cortical areas examined (1, 2).

This finding has been cited well over 500 times according to Google Scholar (March 2012), but, after more than three decades, the conclusion of Rockel et al. remains controversial (3-6) despite a developmental basis for the observations provided by the radial-unit hypothesis and the tangential spread of sister neurons (7, 8). Some researchers claim that the experimental observations of Rockel et al. (1) are technically flawed, whereas others dispute the implication that the structure of neocortex is "the same" across areas and species. The idea of cortical uniformity described by Rockel et al. (1) was first disputed by Haug (1987), who argued that the observations of Rockel et al. could not be confirmed (9); many other critics have surfaced throughout the past 30 y who have denied the cortical uniformity claim of Rockel et al. If Rockel et al. are correct, they have identified a regularity in cortical structure across areas and species that demands a developmental and functional explanation. If the critics of this work are correct, there is no structural regularity to explain.

The "lumpers" among neuroscientists, who like the idea that neocortical areas might have a single canonical circuit, find the Rockel et al. idea of uniformity exciting, whereas the "splitters"—those who believe, at the extreme, that every cortical area in every species is unique—are repelled by it. Recently, a leading neuroanatomist, Pasko Rakic, reported a conversation in which Victor Hamburger commented on the difficulty of disproving incorrect but exciting concepts: "One can spend an entire lifetime correcting a flawed paper published in a reputable journal and still lose the battle if people like the basic idea" (ref. 10, p. 12099). Rakic (2008) continues on to argue that Rockel et al. presented an "exciting concept" (10)—presumably to the many lumpers—and so the idea of cortical uniformity has persisted despite multiple reports that have disputed their findings.

Clearly, the issue of how many neurons are under a square millimeter of neocortical surface is unsettled. Surprisingly, however, the work of Rockel et al. has never been directly replicated. Here, we report a replication of their experiments by counting neurons and glia in neocortex by using accepted stereological methods, and with care to minimize systematic errors: we used only material prepared by our laboratory to minimize variability caused by histological processing procedures, we used counting columns that are oriented perpendicular to the surface of the cortex and within distinct cortical regions, and we studied the species and cortical regions that match the ones Rockel et al. used. The work of Rockel et al. was carried out before stereological procedures were in common use, and complete details about their counting procedures are not presented in the published work. From information given, however, each of two independent observers counted ~750 neurons for each cortical region, a sample that could have been sufficient to provide a good estimate of the number of neurons present under a square millimeter of cortical surface.

Our goal is to evaluate the conclusion of Rockel et al. that the number of neurons underneath a square millimeter of neocortical surface (an idea termed here numerical uniformity) is approximately constant in four cortical areas and in four species Rockel et al. studied: mouse, rat, cat, and monkey. We wished simply to discover if their reported observations are repeatable with modern methods, and do not engage the larger questions relating to structural uniformity across areas and species.

Our data agree with the observations of Rockel et al.: across neocortical regions (excluding primate V1) and across species, we find statistically the same number of neurons underneath a square millimeter of neocortical surface as Rockel et al. reported. We also have extended their initial observations by discovering that the number of glial cells changes systematically with the thickness of the cortex but in a way that is independent of species and cortical area, and we interpret the observation in the context of a theory assuming that astrocytic territories occupy a constant volume of neuropil. In the *Discussion*, we review the literature relevant to the findings of Rockel et al. and attempt to account for the divergences in findings among the various studies.

Results

As noted, our goal in the present work is to reevaluate the conclusions drawn by Rockel et al. by repeating, with more modern methods, their original experiments (1) as closely as possible. The material we have studied differs in three ways from that of Rockel et al. First, because human brains are not available to us, that species has been excluded from the replication. Second, we have excluded primary visual cortex in this report because of fundamental differences between primates and the other species studied; although we have replicated Rockel et al.'s observations on area 17, those observations and extensions of them to other visual areas and primate species will be reported separately. Finally, we have also excluded prefrontal cortex because its small radius of curvature in rodents made it impossible for us to carry out counts perpendicular to the cortical surface

Author contributions: C.N.C. and C.F.S. designed research; C.N.C. performed research; C.F.S. contributed new reagents/analytic tools; C.N.C. analyzed data; and C.N.C. and C.F.S. wrote the paper.

The authors declare no conflict of interest.

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This article contains supporting information online at www.pnas.org/lookup/suppl/doi:10.1073/pnas.1221398110/-/DCSupplemental.

(*Experimental Procedures*). Our detailed analysis presented here thus includes comparisons of 16 neocortical regions from four species and four areas of cortex: the species are mouse, rat, cat, and monkey (Fig. S1 shows histological images), and the areas are motor, somatosensory, parietal, and temporal. We conclude that our observations agree with those first reported by Rockel et al. In addition, we have studied the number of cortical cells that are glia, and find that this number increases systematically with cortical thickness irrespective of species and area.

Cortical Thickness. As an initial comparison of our observations with those of Rockel et al., we report 12 values of cortical thicknesses (four species times three areas) for the same areas and species studied by Rockel et al., and have plotted our values against those published in figures 3–5 of Rockel et al. (1). Comparing the measurements of cortical thickness is important because failure to have counting frames perpendicular to the cortical surface is a major source of systematic error, and comparing thickness measures could reveal this failure. Because Rockel et al. reported the thickness for only three areas (parietal, motor, and primary visual) in their figures, we compare our data for these same areas in our Fig. 1. The slope of the regression line in Fig. 1 is 0.99 (95% CI, 0.87–1.1). Clearly, our measurements concur with those published by Rockel et al., a result expected if the same cortical areas are studied with the same orientation of counting frames relative to the cortical surface.

Cell Numbers and Cortical Thickness. The average total numbers of cells we have counted underneath 1 mm^2 of surface for four cortical regions in four species are presented in Fig. 2*A*. The number of cells (neurons plus glia) under 1 mm^2 of surface is in the range of 100,000 to 150,000 cells, and the regression line has a slope of 18.8 (95% CI, 4.5–32.3) and intercept of 121 (95% CI, 100–142); this fit is significantly different from zero with a probability of $P = 0.01$. The total number of neocortical cells therefore increases systematically with cortical thickness.

An examination of Fig. 2*A* reveals that the cell number variation reflects neocortical thickness rather than species because the cortical thicknesses of the various species are overlapping, and no species has a predominance of points on one or the other side of the regression line. We conclude that the total number of cells underneath 1 mm^2 of neocortical surface is not constant but rather increases systematically with cortical thickness.

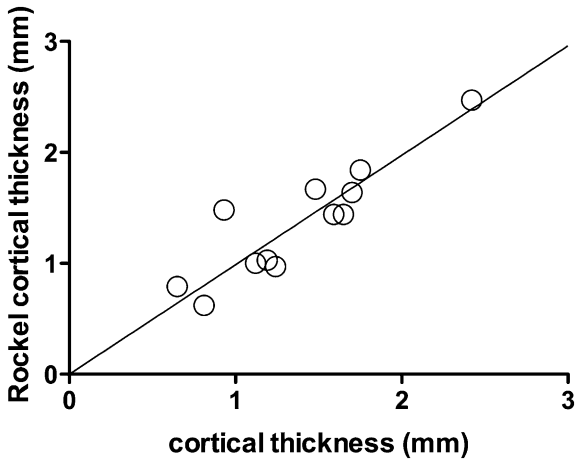


Fig. 1. Cortical thickness measured by Rockel et al. as a function of our measurements of cortical thickness. Data from the four species and three cortical areas as published by Rockel et al. (1980) with our comparable measurements. Regression line has a slope of 0.99 (95% CI, 0.87–1.1).

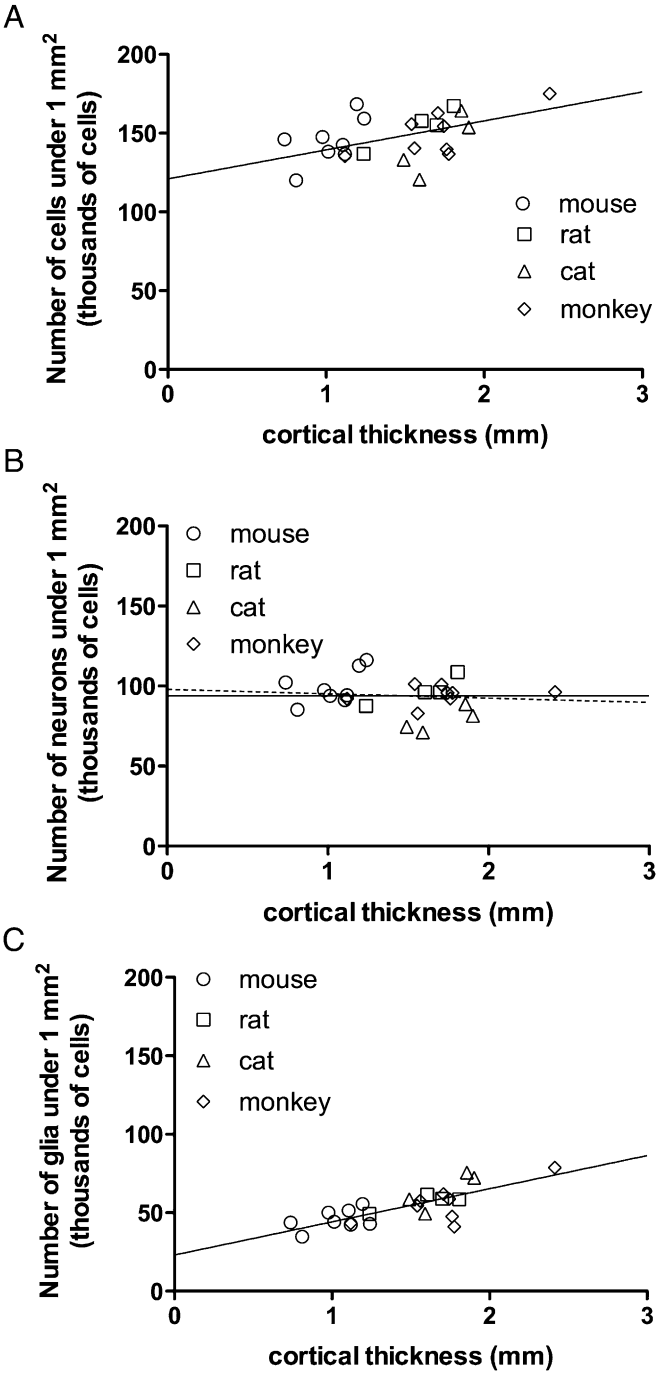


Fig. 2. Cell numbers under 1 mm^2 of cortical surface as a function of cortical thickness. Data presented for four cortical areas and for monkey, cat, rat, and mouse. The abscissa for all three graphs is cortical thickness. (*A*) Number of neurons plus glia under 1 mm^2 of cortical surface. The regression line has a slope of 18.8×10^3 cells per millimeter of cortical thickness and an intercept of 121×10^3 cells. This regression line is significantly different from zero ($P = 0.01$). (*B*) Number of neurons under 1 mm^2 of cortical surface. The regression line has a slope of -2.7×10^3 neurons per millimeter of cortical thickness and an intercept of 94×10^3 neurons. This slope is not significantly different from zero ($P = 0.63$). (*C*) Number of glial cells under 1 mm^2 of cortical surface. The regression line has a slope of 21.1×10^3 glial cells per millimeter of cortical thickness and an intercept of 23×10^3 glial cells. This slope is significantly different from zero ($P = 0.0001$).

What cell types are responsible for this increase in number underneath 1 mm^2 of cortical surface? To answer this question, we plotted neuron number and glial number under 1 mm^2 of

surface as a function of cortical thickness in Fig. 2 *B* and *C*. For neurons, the plot of neuron number as a function of cortical thickness (Fig. 2*B*) is fitted by a regression line with a slope of -2.7×10^3 per millimeter of cortical thickness. This slope is not significantly different from zero ($P = 0.63$), and the mean number of neurons we find underneath a square millimeter of cortical surface is 94.0 ± 10.7 thousand (mean ± SD).

For the same areas and species, Rockel et al. reported 108.9 ± 2.9 thousand neurons. This 14% difference is at least partly accounted for by the fact that Rockel et al. did not correct for shrinkage with their paraffin-embedded material, whereas we studied frozen sections that do not change area in the plane of the section. Frozen sections do, however, shrink perpendicular to the plane of the section, and we have corrected for this by using the cut thickness of sections to determine volume (11). When the number of cells underneath 1 mm^2 of surface in paraffin-embedded material is calculated, the shrinkage will cause an overestimate of the cells counted because the square millimeter will be an underestimate of the actual surface area. We confirm, then, the conclusion drawn by Rockel et al. that the number of neurons underneath 1 mm^2 of neocortical surface is statistically constant for the areas and species we studied. The absolute number of neurons we find is 14% less than the number reported by Rockel et al., and at least part of this difference, perhaps all, results from differences in the methods used (paraffin embedding without shrinkage correction vs. frozen sections).

As is apparent from Fig. 2*C*, the number of glial cells underneath 1 mm^2 of neocortical surface increases with cortical thickness but is unrelated to the species or area studied. The regression line fitted to the Fig. 2*C* data has a slope of 21.1 thousand glial cells per millimeter of cortical thickness, and this slope is significantly different from zero at the $P = 0.0001$ level. We find, then, that glia numbers increase with cortical thickness, and that this increase accounts for the increasing number of all cells underneath 1 mm^2 of neocortical surface, as shown in Fig. 2*A*.

Explanation for Increasing Glia Cell Number with Increasing Cortical Thickness. Astrocytic processes interpenetrate the neuropil, and it has been claimed that each astrocyte has an exclusive territory that supports a collection of synapses, axons, and dendrites (12, 13). We show here that the supported volume per glial cell is constant at 0.047 nL, which is equivalent to a cube of 36 μm per side.

From Fig. 2*C* we know that the number of thousands of glial cells underneath 1 mm^2 of cortical surface is, over the range of cortical thicknesses in our sample, well described by the following linear equation: $N_G = aT + N_0$, where N_G is the number of glia under a square millimeter of cortical surface (in units of number/mm^2); T is the cortical thickness (in units of mm); N_0 is 23×10^3, an additive constant (in units of number/mm^2); and a is 21.1×10^3 (in units of number/mm^3), the slope of the straight line fitted to the data in Fig. 2*C*. If the linear equation is solved for a, it can be seen that this quantity is the density of glial cells (in number of glial cells/mm^3), and that this density is independent of cortical thickness. What this equation means, then, is that, as the cortex becomes thicker, it is adding neuropil volume and glial cells (the number of neurons, as shown earlier, is constant) in a way that keeps glial density constant. The reciprocal of glial density ν is the volume of cortex per glial cell: $\nu = 1/a = 0.047$ nL. That is, each astrocyte is added as neuropil volume is increased by an amount that covers 0.047 nL, which is equivalent to a cube that, as noted earlier, is 36 μm on a side. Thus, the territory of each glial cell is a 36-μm cube. Bushong et al. (2002) and Ogata and Kosaka (2002) report (from reconstructions of filled astrocytes) that the volume of astrocyte territory is equivalent to a cube of approximately 40 μm per side (12, 13). This difference from our 36 μm may result from the interpenetration of neighboring astrocytic territories at their margins.

Discussion

We have concluded that experimental observations of Rockel et al. (1) are, indeed, correct. Furthermore, we find that the number of glial cells per neuron is governed by cortical thickness rather than by species or cortical area. We discuss these two issues in the following sections.

Cortical Thickness as Source of Variable Glial Numbers. Animals with large brains, like primates, generally have more glial cells per neuron than do species with small brains. This observation has led to the idea that there is a "phylogenetic advance" of astrocytes and an increase in the glial cell-to-neuron ratio with brain complexity. For example, *Caenorhabditis elegans* have only a glial cell/neuron ratio of 1/6 (14), whereas the ratio for human cortex is 4 (15), a 24-fold increase. Nedergaard et al. (2003) argued that this increase in the glial/neuron ratio in species with larger brains reflects regulatory functions that glial cells play in networks because the metabolic support provided by glial cells for neurons "are certainly similar among higher vertebrates" (ref. 16, p. 523).

Stolzenburg et al. (1989) studied the glial/neuron ratio in the somatosensory cortex of five insectivores, and found that the ratio scales in an orderly way with cortical thickness, but not with brain weight (17). These authors gave a developmental argument that proposed a mitogenic role for increased potassium ion efflux in a thicker cortex. The observed increase in glia/neuron ratio for these closely related species seems to argue that cortical thickness may be more important than species differences, but the data of Stolzenburg et al. cannot rule out a role for species differences in determining the ratio.

The data presented in Fig. 2*C* show directly that cortical thickness, rather than species, is responsible for an increasing glial/neuron ratio because a thicker cortex in a smaller species can have a higher ratio than a thinner cortex in a larger species. Furthermore, the idea that astrocytes cover the cortical neuropil in a mostly nonoverlapping way, together with the numerical constancy of neurons, can account for the observed glial/neuron ratios.

Other Comparable Studies for Four Species and Four Areas. In addition to the data we have reported and those presented by Rockel et al., we are aware of five other laboratories that have published neuron counts for several of the four species and four areas considered here (4, 5, 18–20). In Fig. 3*A*, we have plotted our data, those of Rockel et al., and data from these other laboratories (20 of 52 data points in Fig. 3 are from the other laboratories). One of these laboratories confirmed Rockel et al.'s numerical uniformity for mouse (18), three laboratories interpreted their data as not supporting Rockel et al.'s conclusions (4, 5, 19), and one laboratory presented values in tables that, combined, gave counts of neurons under 1 mm^2 of cortical surface without discussing Rockel et al.'s claim or comparing the data from the two studies (20). As is apparent from Fig. 3, observations are scattered, with some outliers, but most of the data points fall along a line representing a constant number of neurons per square millimeter of cortical surface. The mean number of neurons under 1 mm^2 of cortical surface is, for all species and areas in Fig. 3, 99,780, with a SD of 13,700, and the regression line in the figure (Fig. 3, dotted line) has a slope that is not significantly different from zero. A cumulative histogram of these data are presented in Fig. 3*B* with a superimposed best-fitting normal distribution function. The fact that these data are well fitted by a normal distribution is consistent with the conclusion that differences between the observations made by various laboratories arise from random counting errors.

Tsai et al. (2009) have also confirmed findings presented in Fig. 3, but the results from their paper have not been included in the figure because of methodological differences (21). Tsai et al. were not concerned with Rockel et al.'s claim of uniformity, but rather used different techniques to address questions related to

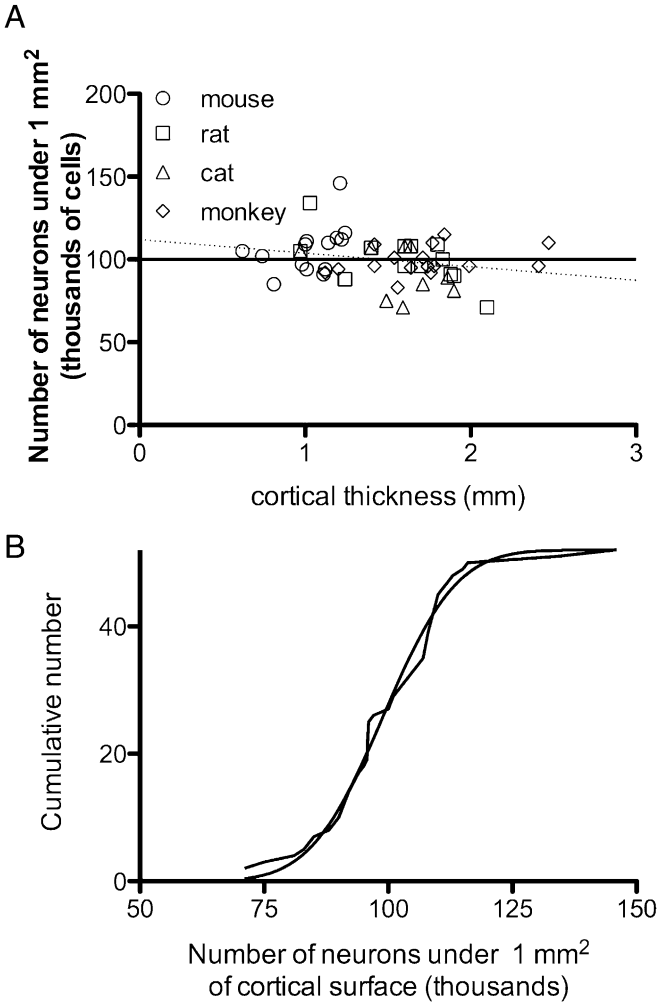


Fig. 3. Summary of data from Rockel et al. (1980), from our work, and from comparable measurements in the literature (see text for references). (*A*) Number of neurons (in thousands) under 1 mm^2 of cortex as a function of cortical thickness. Data from the four species represented are identified by symbols (*Inset*). Regression line (dotted) has a slope of -8.3×10^3 neurons per mm^3 and an intercept of 112 neurons per mm^2. This line is not significantly different from zero. The heavy line represents a constant 97,780 neurons per square millimeter and specifies the mean of all data points displayed. (*B*) Cumulative histogram of neuron number as a function of number of neurons per square millimeter from all the data points in the figure. The smooth curve is a cumulative Gaussian distribution with a mean of 97,780 neurons and an SD of 13,700 neurons (values measured from data points).

cortical vasculature. In the course of their studies, however, they did present neuron counts for four different cortical areas in mouse. Although these researchers found variability of neuron number under 1 mm^2 of mouse cortex, this variability was reported to occur between brain samples and animals, but not between cortical areas (21). Specifically, the paper reports that the number of neurons under 1 mm^2 of cortical surface ranged between 1.025×10^5 and 2.2×10^5, but that the variability was not related to cortical area: the counts for each area varied over the same range, but the mean neuron numbers were not different between areas [figure 10 (21)]. This work had the advantage of counting all the cells present in approximately 1 μL of cortex, but cannot be directly compared with our data presented in Fig. 2 because neurons were discriminated from nonneuronal cells by NeuN immunostaining rather than by histological characteristics. In addition, the same cortical areas were not studied as in Fig. 3, although the studied areas involved motor, somatosensory, parietal, and the transition between sensory and motor cortex. Tsai et al. (21) reported that the average number of neurons under 1 mm^2 of cortical surface is 1.36×10^5, a value approximately one third greater than that found in studies counting cells in histological sections. This difference may be related to the use of NeuN immunostaining to determine if a cell is or is not a neuron (as discussed further in *Four Critical Papers*).

We conclude, then, that the available comparable data in aggregate support the accuracy of Rockel et al.'s original observations. Of course, we cannot exclude differences between the areas and species studied that are too small to be detected in the presence of the noise inherent in neuronal counting by different laboratories.

Four Critical Papers. As illustrated by Fig. 3, data for Rockel et al.'s four areas and four species from various laboratories substantially agree. However, four papers in addition to those considered earlier have used alternative approaches to dispute the accuracy of Rockel et al.'s original observations, and we consider those papers now.

The first paper to question the claim by Rockel et al. of numerical uniformity was that of Haug in 1987 (ref. 9, p. 138): "I think that Rockel's assumption is restricted only to primates, some rodents, and carnivores. It cannot be a general rule for all mammals." This statement was based on data from Haug's laboratory and from the literature, and presents estimates of neuron numbers under 1 mm^2 of cortical surface for approximately 24 species. The numerical values presented range from 19,000 for the elephant to approximately 175,000 for the woolly monkey, an almost 10-fold range. These values were derived from estimates of average cortical volume, average cortical surface area, and average cortical density (neurons per cubic millimeter), with averages computed across multiple cortical areas for each of the various species. Haug divided cortical volume by cortical surface area (to obtain average cortical thickness across different brain areas in millimeters) as a function of brain volume, and then multiplied by his estimates of average neuron density across cortex (measured in thousands of neurons per cubic millimeter) to give thousands of neurons per square millimeter of cortical surface. The estimates of average cortical thickness are quite orderly, but the densities Haug measured are very scattered and account for most of the variability that led Haug to conclude that Rockel et al.'s claim of numerical uniformity is incorrect. Haug provides no discussion of the source of errors in estimates of cortical neuron density.

To evaluate the accuracy of Haug's estimates (9), we turn to another paper that presents data on cortical neuron densities. Tower (1954) also reported data on cortical density as a function of brain volume that overlaps with the species Haug studied, but the Tower data (22) are much less scattered than those presented by Haug. The orderliness of Tower's neuron density data suggests that Haug's scatter could be the result of random measurement errors.

A second problem with Haug's criticism of the findings of Rockel et al. is that Haug makes use of averages across multiple areas of cortical thickness and neuron density (9), whereas an important part of the claim made by Rockel et al. is that the number of neurons under 1 mm^2 of neocortical surface is constant across cortical areas in individual species (except for primate V1) (1). Because we know that differences in cortical thickness and neuron density occur across cortical areas in an individual member of a species, the use of averages for these quantities cannot be assumed to be correct.

To understand the problem, consider the following quantitative argument. The required terminology is as follows: D_{ik} is the neuronal density (average number of neurons per cubic millimeter) of the jth area of the kth species, T_{jk} is the thickness (in millimeters) of the jth area and k^{th} species, and N_{jk} is equal to $D_{ik}T_{jk}$, the number of neurons under 1 mm^2 of cortical surface

for the jth area and k^{th} species. Furthermore, define average values of D_k as equal to $\langle D_{ik} \rangle_I$, T_k as equal to $\langle T_{ik} \rangle_j$, and N_k as equal to $\langle N_{jk} \rangle_j$ (the averaging operation, basically a sum over areas, is denoted by the angle brackets) across j^{th} areas in the kth species for each of these quantities. Because the sum of products is not, in general, equal to the product of sums, even if the N_k values are constant, the quantities $D_k T_k$ will always be greater than N_k if cortical thickness and neuron density vary between areas and species.

Furthermore, if primates are included [as was the case for Haug (9)], averages across neocortical areas are particularly inappropriate because the primary visual cortex cannot, according to Rockel et al., be treated in the same way as the other areas.

In summary, Haug's criticisms of the Rockel et al. conclusions cannot be sustained without assurances that his cortical densities are accurate and that the use of the average across multiple areas of densities and cortical thickness is justified.

Prothero (1997) had a somewhat different criticism of Rockel et al. (23). His point was that Rockel et al. had taken an inadequate sample of neuron number under 1 mm^2 of cortex because the column of cells sampled was too small in cross-section. Prothero terms the measurement made by Rockel et al. a "line count" (i.e., the number of cells across the cortical thickness estimated from a thin column of cortex) and states (ref. 23, p. 514) that Rockel et al. "extrapolated from an invariant line count to an invariant neuron density, computed per area of cortical surface."

In fact, Rockel et al. took an adequate sample of the number of neurons under 1 mm^2 of cortical surface for each area (between 1% and 2% of the total neuron number) (1), as did we, to provide a statistically accurate sample of neuron number under 1 mm^2 of cortical surface. Although Rockel et al. did not detail their counting methods, the fact that our independent replication of their work produced the same numerical estimates provides support for the idea that Rockel et al. did, in fact, provide an adequate sample.

Herculano-Houzel and Lent (2008) have also disputed Rockel et al.'s conclusion of numerical uniformity based on cell counts with the isotropic fractionator method and measurements of average cortical volume and surface area (6). This work is, however, subject to the same criticisms outlined earlier: the use of average values across cortical areas of primates is unjustified, based on Rockel et al.'s observation that at least one area (V1) has a different number of neurons under 1 mm^2 of cortex. Furthermore, Collins et al. (2010) have reported that visual areas other than V1 have increased numbers of neurons (24). These observations stress the difficulty of interpreting cell counts that are averages across all cortex.

In addition, the interpretation of data obtained with the isotropic fractionator method is made more difficult by two main technical difficulties to which this method is subject. The first has to do with mechanical separation of cell nuclei and the second with the use of NeuN immunostaining to distinguish between neurons and other nonneuronal cell types. These difficulties are considered in turn in the next two paragraphs.

The isotropic fractionator method depends on the mechanical disaggregation of neuronal tissue that thoroughly separates all cell nuclei without destroying any (25). If the disaggregation is incomplete, cell clumps will remain and will cause the cell number to be underestimated. At the same time, if the mechanical disaggregation is too stringent, nuclei will be destroyed, and the cell number will again be underestimated. Some independent check on the cell number—DNA extraction and quantification, for example—is necessary to verify that the estimated cell number is correct.

Although NeuN will stain most neurons selectively (26, 27), the fluorescence associated with the presence of this antibody is not all-or-none but rather is graded from neurons to nonneurons, i.e., glia. Examples of this grading are given in figure 8c of Tsai et al. (2009) (21) and in figures S2 and S3 of Collins et al. (2010), in which histograms of the number of cells associated with each fluorescence level are presented (28). As is clear from the two figures of Collins et al., there is overlap between the fluorescence intensity of neuronal and nonneuronal cells. Until the fluorescence threshold for accurate separation of neurons from glia is verified, this overlap in staining of cell types is a source of error of unknown magnitude.

In summary, using quantities averaged across cortical areas is an inaccurate way to determine the limits of validity of Rockel et al.'s assertion of neuronal uniformity, and the magnitude of errors caused by using NeuN to classify cells as neurons and glia are not known.

Rockel et al. presented a particular view of cortical uniformity that divided cortical areas into two categories, the "usual" types and the single special type (V1 in primates). To evaluate and extend this view, accurate measures of neuronal numbers must be made for more areas and species, a program initiated by Collins et al. (2010) (24). These researchers have used the isotropic fractionator method for flattened primate cortex, and find a picture that appears to be more complicated than the view presented by Rockel et al. The problem with this initial step has been uncertainties about the error magnitudes associated with the method used; without determining error magnitudes, relating these data to the Rockel et al. picture is difficult. As techniques improve, particularly methods like imaging all cells present as done by Tsai et al. (2009) (21), we can anticipate that the Rockel et al. model will be confirmed and/or extended.

In summary, then, the data directly comparable from various laboratories, in aggregate, supports the original claim by Rockel et al. (1) of numerical neuronal uniformity in the four areas and four species considered here. Further, the criticisms of Rockel et al.'s conclusions raised by the four papers discussed here earlier cannot be taken as definitive until the uncertainties noted for those studies are addressed.

Experimental Procedures

Overview. Our main goal was to repeat experiments conducted by Rockel et al. (1); thus, we collected neuronal counts with the same methodological principles in their original paper; counts were taken in column volumes from the pial surface to the white matter layer, and the same brain regions in similar species were used. The one difference between Rockel et al.'s methods and ours is that we conducted counts by using unbiased stereological methods; failure to use these modern methods has been one strong criticism of the work of Rockel et al.

Histological Procedures. Brain tissues from the mouse, rat, cat, and rhesus macaque were analyzed. Animal care protocols were approved by the Salk Institute Animal and Use Committee and conform to US Department of Agriculture regulations and National Institutes of Health guidelines for humane care and use of laboratory animals. Each specimen was perfused with aldehyde fixative agents and stored long-term in 10% formalin. In preparation for cutting, brains were submerged in solutions of 10% (wt/vol) glycerol and 10% formalin until the brain sank, and then moved into 20% glycerol and 10% formalin until the brain sank again; the average time in each solution was 3 to 10 d. These cryoprotected brains were then cut coronally on a freezing microtome at a thickness of 40 or 50 μm. Every sixth or 12th section was stained with thionin for visualization of Nissl bodies. Specific characteristics of all specimens are detailed in Table S1.

Data Collection. Only left hemispheres were analyzed, and each region's gray matter was outlined, based on standard atlases and primary literature, in Neurolucida software (version 7.52; MBF Bioscience) at low magnification (2× objective). In mouse, rat, cat, and rhesus monkey, the following regions were outlined: frontal cortex, primary motor cortex, parietal cortex, primary somatosensory cortex, temporal cortex, and primary visual cortex. In addition to V1, other visual cortical areas were counted in the brown capuchin, crab-eating macaque, and rhesus monkey, including secondary visual area, visual area 3, visual area 4, and visual area 5 or middle temporal area.

Ten- or 20-μm-wide columns, perpendicular to the pial surface, extending down to the white matter, were delineated by using the contours option, and their width was confirmed by using the quick measure line option in

Neurolucida. Column position was determined based on its estimated perpendicularity to our coronal cut and the surface of the brain (Fig. 1). Neurons and glia were then counted in adjacent Nissl-stained sections at 100× oil magnification in approximately three or four sections per region in each specimen. The counting column functioned as a typical disector whereby neurons, identified by the presence of a nucleolus, were marked as they come into focus given that they also fell within the acceptance lines of the disector (29-31). Approximately 50 to 100 objects of interest were counted in each 3D column, and eight columns were counted per region. The total number of neurons counted ranged from 400 to 800 neurons per specimen for each region.

We counted all the neurons and glia in our column from the pial surface to the white matter without the use of guard zones. A detailed discussion of guard zone use as it pertains to stereological methods and data collection for frozen sections was reported by Carlo and Stevens (2011) (11).

Calculations and Statistical Analyses. Standard least-squares regression analyses were performed on $\log_{10}$ transformed data in Prism5 statistical software (GraphPad Software). All regression residuals were analyzed for normality by using the Shapiro–Wilk test statistic ($W < 0.05$ to reject the null hypothesis that the data follow a normal distribution).

Controls for Systematic Errors. The experiments described here are potentially subject to two types of systematic errors. The first arises when the data for a particular area and species are inhomogeneous. This could happen, for example, if data thought to come from a single cortical area actually came from two different areas as a result of areal misidentification in some sections. The second type of error would arise if the cell counts were carried out in counting columns that are not perpendicular to the cortical surface; if this occurred, overestimates of cell number would result, and it would appear that various cortical areas had different numbers of neurons under 1 mm^2 of cortical surface when, in fact, they did not. The controls for both sorts of systematic errors are described here in turn.

Random fluctuations in the number of cells counted from one counting column to the next will occur, but one would expect these fluctuations to follow a Poisson distribution if all columns have homogeneous properties. In this case, a Poisson distribution should be approximated by a Gaussian distribution whose variance is equal to the mean number of cells counted for that cortical area. For each cortical area and each species studied, we constructed a cumulative histogram of fraction of the total number of cells counted in each counting column, and tested (with a Kolmogorov–Smirnov test) to see if the histograms were satisfactorily fitted by the Gaussian distribution whose SD is equal to the square root of the mean number of cells counted for that cortical area. If regions with inhomogeneous properties were grouped together, the actual SD would be greater than the one predicted. In all cases, the predicted cumulative Gaussian distribution was not significantly different from the observed histogram.

If counting columns are not perpendicular to the cortical surface, longer counting columns will have more cells than shorter ones. Cortex naturally varies in depth from place to place even within a single cortical area, and so the lengths of all counting columns are not exactly equal. According to Rockel et al., the number of neurons per counting column should be independent of these statistical fluctuations in column length. To determine if columns of different length conformed to Rockel et al.'s prediction or to what would be seen if the different column length arose from nonperpendicular columns, we plotted the number of neurons counted as a function of the cortical thickness (i.e., length of the counting column), and determined if the regression line fitted to these data had a slope significantly different from zero. None of the counts for the four areas and four species exhibited a dependence of neuron number on column length.

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